Bayesian Typhoon Track Progression Specification

Purpose

This specification defines a Bayesian progression model for advancing a typhoon through time. The objective is to provide the coding team with a practical probabilistic architecture for simulating forward typhoon motion across a time horizon, with uncertainty represented explicitly rather than through a single deterministic path.[cite:102][cite:105][cite:106][cite:107]

The model is intended for a simplified simulation framework rather than a definitive operational forecast system. It should therefore be interpretable, modular, easy to calibrate, and compatible with the existing storm abstractions already present in the codebase, especially the time-indexed track representation, storm speed metadata, and footprint logic.[cite:98][cite:99][cite:100]

Objective

The model should answer the following question:

Given the storm state at time $t - 1$, and any environmental or observational information available at time t, what is the posterior distribution of the typhoon state at time t?[cite:105][cite:106][cite:107]

The model should produce a probabilistic sequence of storm center locations, headings, forward speeds, peak winds, and size metrics over the chosen time horizon. It should support both free simulation and sequential updating when new information becomes available.[cite:107][cite:114]

Design principles

The Bayesian progression model should satisfy the following principles.

1. Storm movement should be represented as a stochastic state process, not a fixed line segment between start and end points.[cite:107][cite:114]
2. The model should update the storm state sequentially over time using Bayes' rule.[cite:105][cite:107]
3. Forward motion should be driven by physically meaningful state variables, including position, heading, translation speed, wind intensity, and storm size.[cite:98][cite:99]
4. The model should allow uncertainty in both motion and structure, not only in one-dimensional track position.[cite:107][cite:114]
5. The first release should prioritize interpretability and calibration over full meteorological completeness.[cite:106][cite:112]

Model overview

The recommended design is a Bayesian state-space model. The typhoon is described by an unobserved latent state that evolves over time, while optional data or soft constraints provide likelihood information for updating the posterior state distribution.[cite:105][cite:107][cite:114]

At each time step the workflow is:

1. Start with the posterior distribution from the previous step.
2. Propagate that state forward using a transition model.
3. Evaluate any new information or plausibility constraints.
4. Update the state distribution using Bayes' rule.
5. Continue until the required time horizon is reached.[cite:105][cite:107]

State definition

The latent state at time t should be defined as:

$$s_t = (\lambda_t, \phi_t, u_t, \psi_t, V_t, R_t, k_t)$$

Equation (1)

Where:

- λ_t is longitude.
- ϕ_t is latitude.
- u_t is translation speed.
- ψ_t is heading or motion bearing.
- V_t is maximum sustained wind at the storm center.
- R_t is a size vector, at minimum including radius of maximum wind and outer wind radius.
- k_t is an optional track regime or hidden motion class.[cite:98][cite:99][cite:100][cite:106]

A more explicit size vector may be defined as:

$$R_t = (R_{\max,t}, R_{\text{outer},t})$$

Equation (2)

The state should be serializable and compatible with the typhoon event model proposed in the wind-field specification.[cite:98]

Regime variable

A discrete regime variable should be included because Bayesian typhoon work often benefits from track-pattern categorization. Historical storms tend to fall into recurring path families, and modeling motion conditional on regime is usually more stable and interpretable than using one global motion rule.[cite:106][cite:109][cite:115]

Suggested regimes include:

- Straight westward track.
- Northwest-moving recurver.
- Sharp recurvature track.
- Slow or stalled track.
- Landfall decay track.[cite:106][cite:109]

The regime may be fixed for an event or may evolve with low transition probability over time. The simplest first version should sample it once at storm genesis and hold it fixed unless evidence strongly suggests a transition.[cite:106][cite:109]

Bayesian formulation

The sequential Bayesian update is:

$$p(s_t \mid y_{1:t}) \propto p(y_t \mid s_t)\, p(s_t \mid y_{1:t-1})$$

Equation (3)

Where:

- s_t is the latent state at time t.
- y_t is information available at time t, including observations, environmental guidance, or synthetic plausibility constraints.
- $p(s_t \mid y_{1:t-1})$ is the predictive prior from the transition model.[cite:105][cite:107]

The predictive prior is obtained from:

$$p(s_t \mid y_{1:t-1}) = \int p(s_t \mid s_{t-1})\, p(s_{t-1} \mid y_{1:t-1})\, ds_{t-1}$$

Equation (4)

This is the canonical state-space formulation and should be the conceptual basis for the implementation.[cite:107][cite:114]

Transition model

The transition model governs how the typhoon moves from one step to the next. It should update motion, wind, and size jointly, or at least in coupled blocks.

Motion update

The recommended first-order motion block is:

$$u_t \sim \mathcal{N}(\mu_u(s_{t-1}, z_t, k_t), \sigma_u^2(k_t))$$

$$\psi_t \sim \mathcal{N}(\mu_\psi(s_{t-1}, z_t, k_t), \sigma_\psi^2(k_t))$$

Equation (5)

Where:

- z_t is environmental forcing or steering information.
- k_t is the regime variable.[cite:106][cite:107][cite:114]

The conditional means should depend on at least:

- Previous speed and heading.
- Position and latitude.
- Regime class.
- Land or ocean interaction.
- Recurvature tendency at higher latitudes.[cite:106][cite:109][cite:112]

Position update

Given translation speed and heading, the center should be advanced using a local advection rule. In planar approximation:

$$\Delta x_t = u_t \Delta t \cos(\psi_t), \qquad \Delta y_t = u_t \Delta t \sin(\psi_t)$$

Equation (6)

And then:

$$\lambda_t = \lambda_{t-1} + c_\lambda \Delta x_t, \qquad \phi_t = \phi_{t-1} + c_\phi \Delta y_t$$

Equation (7)

Where c_λ and c_ϕ convert km displacement into longitude and latitude increments. The implementation may use Haversine or geodesic utilities for better geographic fidelity, as the current code already includes distance handling logic through Haversine calculations.[cite:99][cite:100]

Wind update

The wind state should evolve with persistence and environmental moderation:

$$V_t \sim \mathcal{N}(\mu_V(V_{t-1}, z_t, k_t), \sigma_V^2)$$

Equation (8)

The mean function may include:

- Persistence from previous wind.
- Intensification tendency over warm water.
- Weakening due to land interaction.
- Optional decay after recurvature or extratropical transition.

A simple landfall weakening rule may be:

$$V_t = V_{t-1} \exp(-k_{\text{land}} \Delta t)$$

Equation (9)

When the storm is over land or land fraction exceeds a threshold.[cite:100]

Size update

The storm size state should evolve stochastically but smoothly:

$$R_{\max,t} \sim \mathcal{N}(\mu_{R_{\max}}(R_{\max,t-1}, V_t, z_t), \sigma^2_{R_{\max}})$$

$$R_{\text{outer},t} \sim \mathcal{N}(\mu_{R_{\text{outer}}}(R_{\text{outer},t-1}, V_t, z_t), \sigma^2_{R_{\text{outer}}})$$

Equation (10)

This keeps the wind-field model structurally coupled to the motion model without requiring a full pressure-wind system.[cite:98][cite:99]

Regime-conditioned mixture model

The transition model should be regime-conditioned. Formally:

$$p(s_t \mid s_{t-1}) = \sum_k p(s_t \mid s_{t-1}, k)\, p(k \mid s_{t-1})$$

Equation (11)

In the simplest implementation, the event is assigned a regime at genesis:

$$k \sim \text{Categorical}(\pi_1, \ldots, \pi_K)$$

Equation (12)

Where the prior weights π_k depend on genesis location, season, and perhaps initial heading.[cite:106][cite:109][cite:115]

This is strongly recommended because it provides a principled Bayesian way to encode broad path behavior without forcing the coding team into a highly nonlinear global model.[cite:106][cite:109]

Likelihood model

The likelihood $p(y_t \mid s_t)$ may be fed by one of three sources.

1. Real observations

If real-time observations are available, the likelihood may use observed storm center positions, observed headings, forecast advisory data, or wind estimates.[cite:105][cite:107]

2. External guidance

If external model guidance exists, the likelihood may represent agreement with a forecast ensemble or consensus track, following a Bayesian model averaging style approach.[cite:105]

3. Soft constraints for simulation mode

If the model is running as a synthetic simulator, the likelihood may be replaced by plausibility scores. Examples include:

- Penalize unrealistically sharp heading changes.
- Penalize implausible jumps in translation speed.
- Penalize motion over terrain inconsistent with the chosen regime.
- Penalize trajectories that violate basin boundaries or historical behavior.

This allows the same Bayesian machinery to be used even when there are no real observations to assimilate.[cite:106][cite:112]

Inference method

The recommended implementation method is Sequential Monte Carlo, also known as a particle filter. This is the most natural way to represent uncertainty in a nonlinear storm state process.[cite:107]

Particle representation

At time t, represent the posterior as a weighted set of particles:

$$p(s_t \mid y_{1:t}) \approx \sum_{i=1}^{N} w_t^{(i)}\, \delta(s_t - s_t^{(i)})$$

Equation (13)

Where:

- $s_t^{(i)}$ is particle i.
- $w_t^{(i)}$ is its normalized weight.[cite:107]

Particle filtering workflow

1. Initialize particles from the genesis prior.
2. Propagate each particle using the transition model.
3. Compute weights from the likelihood or plausibility function.
4. Normalize weights.
5. Resample if the effective sample size falls below a threshold.
6. Repeat for the next time step.[cite:107]

This should be the default implementation. A deterministic mean-path fallback may be added later, but the probabilistic particle version should be considered the canonical Bayesian model.[cite:107]

Initialization

At genesis time $t = 0$, the prior should define distributions for:

- Genesis longitude and latitude.
- Initial heading.
- Initial translation speed.
- Initial maximum wind.
- Initial size.
- Regime class.[cite:78][cite:79][cite:100][cite:106]

The current distribution module already provides a scenario-family concept with adjustable tail thickness.[cite:78][cite:79] That concept should be retained and reinterpreted so that peak wind, duration, and perhaps initial translation speed can be sampled from scenario-conditioned priors.[cite:78][cite:79]

Environmental forcing

The state transition functions may accept an environmental vector z_t. The first version may keep this simple.

Recommended optional forcing variables:

- Latitude band or recurvature proxy.
- Ocean versus land flag.
- Basin-specific steering tendency.
- Seasonal modifier.
- Optional shear or SST proxy.

If environmental data are unavailable, the model may still work by using regime-conditioned priors plus position-based heuristics.[cite:106][cite:112]

Output requirements

The Bayesian progression model should produce the following outputs.

Posterior trajectory outputs

- Mean or median storm center path.
- Credible bands or confidence envelopes for location over time.
- Ensemble of particle tracks.
- Posterior distributions for heading and translation speed.[cite:105][cite:107]

Wind-state outputs

- Posterior mean and quantiles for maximum wind over time.
- Posterior mean and quantiles for storm size metrics over time.
- Optional scenario-conditioned severity summaries.[cite:98][cite:99]

Event summaries

- Probability of landfall in each region.
- Probability of recurvature.
- Probability of exceeding wind thresholds at chosen horizons.
- Time-to-landfall or time-to-closest-approach distributions.[cite:105][cite:107]

Interface with the typhoon wind-field model

The Bayesian progression model should feed directly into the typhoon wind-field model from the earlier specification. Each posterior particle or posterior mean state at time t should be usable as an input to the radial wind-field module, which then computes location-specific winds.[cite:98][cite:99]

The interface contract should therefore expose, at each time step:

- Center position.
- Translation speed.
- Heading.
- Maximum sustained wind.
- Radius of maximum wind.
- Outer wind radius.
- Land interaction flags.

This ensures the motion model and the wind-field model remain decoupled but interoperable.[cite:98][cite:99]

Required new models and modules

Module	Purpose	Notes
`BayesianTyphoonState`	Canonical latent state container	Should include motion, wind, size, and regime.[cite:98]
`TrackRegimeModel`	Regime priors and regime-conditioned transition parameters	Strongly recommended for interpretability.[cite:106][cite:109]
`MotionTransitionModel`	Updates heading and translation speed	Core of forward progression.[cite:107][cite:114]
`IntensityTransitionModel`	Updates maximum wind through time	Should support land weakening.[cite:100]
`SizeTransitionModel`	Updates wind radii through time	Needed by wind-field model.[cite:98][cite:99]

Module	Purpose	Notes
`ObservationLikelihoodModel`	Scores particles against observations or plausibility constraints	Must support observation-free simulation mode.[cite:105][cite:107]
`ParticleFilterEngine`	Sequential Monte Carlo implementation	Recommended default inference engine.[cite:107]
`PosteriorExporter`	Writes trajectories, credible regions, and summaries	Output-facing module

Refactoring guidance for current codebase

`storm_factory.py`

The current storm factory builds deterministic tracks and intensity profiles.[cite:100] It should be refactored into a prior generator that samples an initial state, regime, and track skeleton rather than a fully fixed event.[cite:100][cite:106]

`data_structures.py`

The current data structures are event-oriented and use generic intensity fields.[cite:98] They should be extended to support a Bayesian state object, regime variables, posterior particles, and explicit wind-based semantics.[cite:98]

`forward_model.py`

The current forward model interpolates deterministic state along a track and computes local decay from the center.[cite:99] This logic should remain available, but it should become the downstream consumer of posterior state sequences produced by the Bayesian progression engine.[cite:99]

Calibration requirements

Calibration should proceed in layers.

1. Regime identification from historical track families.[cite:106][cite:109][cite:115]
2. Conditional motion parameter fitting by regime for heading and speed updates.[cite:106][cite:107]
3. Wind persistence and land decay fitting.[cite:100]
4. Size evolution fitting, at least through heuristic rules initially.[cite:98][cite:99]
5. Likelihood function tuning for observation-free simulation plausibility.[cite:106][cite:112]

All parameters should be externalized into configuration files or parameter objects rather than embedded in code constants.[cite:79]

Validation requirements

The coding team should validate the Bayesian progression model against the following criteria.

- Plausible particle-track spread through time, widening with forecast horizon.[cite:105][cite:107]
- Realistic smoothness of heading and speed paths.[cite:106][cite:112]
- Reasonable distribution of recurvature and landfall behavior by regime.[cite:106][cite:109]
- Stable posterior behavior under different particle counts and time steps.[cite:107]
- Consistency of progression outputs with the downstream wind-field footprint generator.[cite:99]

Non-goals

The following are out of scope for the first implementation:

- Full atmospheric data assimilation.
- Full dynamical numerical weather prediction.
- Joint ocean-atmosphere coupling.
- Full pressure-wind structural models.
- Operational forecast-grade verification pipelines.[cite:107][cite:112]

The focus should remain on a reduced-form Bayesian progression engine suitable for simulation, stress testing, and exploratory wind-risk modelling.[cite:106][cite:107]

Implementation summary

The coding team should implement typhoon forward movement as a Bayesian sequential state-space process with regime-conditioned motion, stochastic updates to speed and heading, coupled updates to wind and size, and inference performed through a particle filter.[cite:105][cite:106][cite:107][cite:114]

This is the recommended minimum viable Bayesian architecture because it captures uncertainty explicitly, remains interpretable, aligns with published Bayesian tropical cyclone track approaches, and integrates naturally with the existing track-based storm abstractions already present in the codebase.[cite:98][cite:99][cite:100][cite:106][cite:107]

Tail events and intensity skew

The Bayesian typhoon progression framework must explicitly represent tail events, meaning low-probability but high-severity typhoon realizations. In this specification, a tail event is not a separate post-storm physical phase. Instead, it is an extreme realization drawn from the same probabilistic lifecycle model, lying in the far right tail of one or more state dimensions such as peak wind, storm size, duration, translation behavior, or track rarity.[cite:3][cite:27][cite:123]

This distinction is important for architecture. The model should not create a separate “tail-event engine.” It should instead generate tail events naturally from skewed or fat-tailed prior and transition distributions, so that extreme storms remain consistent with the same Bayesian state-space machinery used for ordinary events.[cite:106][cite:107][cite:123]

Tail-event dimensions

Tail behavior should be represented in at least five dimensions.

1. **Intensity tail**: unusually high peak sustained wind, including the equivalent of the most severe typhoon classes in the chosen basin.[cite:27][cite:126]
2. **Size tail**: unusually large radius of maximum wind or outer wind radius, creating broad damaging-wind footprints even when peak wind is not uniquely extreme.[cite:27][cite:126]
3. **Duration tail**: unusually long-lived storms or storms that maintain damaging winds over long periods.[cite:3][cite:27]
4. **Track tail**: low-probability path geometries, such as unusual recurvature, unexpected landfall corridors, or tracks that remain close to exposed coastlines for extended periods.[cite:106][cite:109][cite:123]
5. **Compound tail**: combinations of high intensity, large size, slow translation, and vulnerable landfall geometry occurring together.[cite:123][cite:126]

The first implementation may focus mainly on intensity, size, and track tails, but the interfaces should reserve capacity for compound tail scenarios because these often dominate practical risk outcomes.[cite:123][cite:126]

Intensity skew requirement

Peak wind intensity should not be modeled with a symmetric light-tailed distribution. The codebase already contains a hybrid distribution concept in which ordinary values are modeled with a truncated normal structure and extreme values are modeled with a Pareto tail.[cite:78][cite:79] That logic should be retained and reinterpreted as a peak-wind severity distribution.

The exceedance distribution for peak wind V_{peak} should be defined as:

$$P(V > v) = \begin{cases} 1 - \Phi\left(\dfrac{v-\mu}{\sigma}\right), & v \le v_T \\ P(V > v_T)\left(\dfrac{v_T}{v}\right)^{\alpha}, & v > v_T \end{cases}$$

Equation (14)

Where:

- μ is the baseline mean peak wind.
- σ is the baseline standard deviation.
- v_T is the threshold above which tail behavior dominates.
- α is the Pareto tail index, with smaller values representing fatter tails.[cite:78][cite:79]

This specification requires the coding team to preserve positive skew in the peak-wind distribution. That skew is the mechanism by which the model can generate rare but plausible extreme typhoons for stress testing and tail-risk analysis.[cite:78][cite:79][cite:123]

Tail-event classification

The model should support explicit tail-event tagging. A typhoon realization should be classified as a tail event if one or more calibrated state variables exceed predefined quantile thresholds under the reference climate or scenario family.

A recommended first version is:

- **Moderate tail event**: any of V_{peak}, R_{outer}, event duration, or landfall exposure metric exceeds the 95th percentile.[cite:78][cite:79]
- **Severe tail event**: any key metric exceeds the 99th percentile, or a joint-event score exceeds its 97.5th percentile.[cite:78][cite:79]
- **Compound tail event**: two or more tail dimensions simultaneously exceed chosen high-quantile thresholds.[cite:123][cite:126]

The quantiles should be configurable by basin and scenario family, not hard-coded globally.[cite:79]

Joint tail score

To support risk ranking and filtering, the model should optionally compute a joint tail severity score. A practical first formulation is:

$$S_{\text{tail}} = w_V q_V + w_R q_R + w_D q_D + w_T q_T + w_C q_C$$

Equation (15)

Where:

- q_V is the normalized upper-tail score for peak wind.
- q_R is the normalized upper-tail score for size.
- q_D is the normalized upper-tail score for duration.
- q_T is the normalized rarity score for track geometry.
- q_C is the normalized coastal or landfall exposure score.
- w_* are configurable weights.

This score is not a substitute for the underlying probabilistic model. It is a convenience metric for identifying and ranking extreme realizations drawn from the posterior or scenario ensemble.[cite:123][cite:126]

Tail behavior in Bayesian progression

Tail events should emerge naturally in the Bayesian progression model through three mechanisms.

1. **Tail-prior sampling**: initial peak wind, size, or duration are sampled from skewed prior distributions with fat-tail support.[cite:78][cite:79]
2. **Regime-conditioned rare tracks**: low-probability regimes or low-probability path updates are retained with non-zero posterior mass, allowing unusual but plausible trajectories.[cite:106][cite:109]
3. **Persistence of extremes**: high wind, large size, or slow translation should have persistence structure, so once an event enters an extreme state it does not collapse unrealistically in the next time step unless land or

adverse forcing justifies that change.[cite:100][cite:107]

This means the transition model should not be too mean-reverting in the extreme tail. It should allow rare events to remain extreme over multiple steps when environmental conditions support persistence.[cite:107][cite:123]

Stress-scenario families

The scenario-family design already present in the codebase should be preserved and extended for typhoon tails.[cite:79] At minimum, the following family types should be supported:

Scenario family	Intended effect on tail behavior
Historical	Matches observed or reference-climate tail thickness.[cite:79]
Baseline	Slightly conservative central estimate with realistic upper tail.[cite:79]
Moderate stress	Higher mean peak wind, lower tail threshold, moderately fatter tail.[cite:79]
Severe stress	Higher mean, stronger persistence, fatter extreme tail, more frequent large storms.[cite:79]
Extreme	Strong positive skew, very fat upper tail, elevated probability of compound extremes.[cite:79]

In implementation terms, scenario families should influence at least:

- Peak-wind distribution parameters μ, σ, v_T, α.[cite:79]
- Size distribution parameters for $R_{\max}$ and R_{outer}.
- Regime priors for rare track behavior.[cite:106][cite:109]
- Persistence or decay parameters in the transition model.

Tail outputs

The model should produce explicit outputs for tail-event analysis. These should include:

- Tail classification flag.
- Quantile rank for peak wind, size, duration, and track rarity.
- Joint tail score.[cite:123][cite:126]
- Scenario-family identifier.[cite:79]
- Exceedance probabilities for user-defined severity thresholds.[cite:78][cite:79]

These outputs should be available both at event level and, where relevant, at location level after coupling to the wind-field module.[cite:98][cite:99]

Validation of tail behavior

The coding team should validate not only mean behavior but also the upper tail. Validation should include:

- Stability of high quantiles under repeated Monte Carlo simulation.[cite:78][cite:79]
- Sensitivity of tail frequency to the Pareto index and threshold.[cite:78][cite:79]
- Plausibility of rare track realizations under regime priors.[cite:106][cite:109]

- Persistence of extreme states through time under favorable conditions.[cite:107]
- Sensible separation between ordinary, severe, and compound tail scenarios.[cite:123][cite:126]

Implementation note

The coding team should treat tail events as first-class outputs of the model rather than ad hoc exceptions. The architecture should therefore expose the skew and fat-tail settings explicitly in configuration, scenario definitions, posterior summaries, and event metadata.[cite:79][cite:123]

Hierarchical joint hazard model for wind, surge, and rain

The Bayesian typhoon progression model should be extended with a hierarchical multivariate hazard layer for wind, surge, and rain. This is the recommended architecture for handling their dependence structure because most of the correlation between these variables should arise from common storm-level drivers rather than from a single undifferentiated correlation matrix.[cite:140][cite:141][cite:143][cite:145]

This specification considers the hierarchical approach to be at a sufficient level of sophistication for the intended platform. It is materially more realistic than a static Gaussian correlation model, but it remains modular, calibratable, and explainable enough for implementation, review, and model governance.[cite:145][cite:148][cite:153]

Conceptual structure

The model should be organized into three layers.

1. **Latent storm-state layer**: a Bayesian state-space representation of the storm, including position, heading, translation speed, peak wind, size, land interaction, and other core storm descriptors.[cite:98][cite:99][cite:100]
2. **Hazard-response layer**: conditional models for local wind, surge, and rain given the latent storm state.[cite:143][cite:149][cite:150]
3. **Residual dependence layer**: a tail-aware dependence model for the remaining co-movement not explained by the latent storm state.[cite:145][cite:151]

This decomposition is the preferred implementation because it aligns with physical reasoning. Wind, surge, and rain are not independent, but they are not jointly driven by an arbitrary covariance structure either. They are linked because the same underlying storm state produces all three hazards.[cite:141][cite:143][cite:147]

Joint distribution

The joint hazard structure should be written conceptually as:

$$P(W, S, R) = \int P(W \mid Z)\, P(S \mid Z)\, P(R \mid Z)\, C_{\text{res}}(W, S, R \mid Z)\, P(Z)\, dZ$$

Equation (16)

Where:

- Z is the latent storm state.

- W is local wind.
- S is local surge or surge proxy.
- R is local rainfall or rainfall proxy.
- C_{res} is the residual dependence structure.[cite:145][cite:148][cite:153]

The coding team should interpret this as follows. Most cross-variable dependence should be induced by the common storm state Z. The residual dependence term should only capture the dependence that remains after conditioning on Z, especially in the upper tail.[cite:145][cite:151]

Latent storm-state layer

The storm-state layer is already defined in the Bayesian progression model. For the joint hazard model, the key latent variables should include at least:

- Center location.
- Heading.
- Translation speed.
- Peak sustained wind.
- Radius of maximum wind.
- Outer wind radius.
- Land interaction indicator.
- Optional pressure deficit proxy.
- Optional moisture or convective efficiency proxy.[cite:98][cite:99][cite:100]

These variables should be treated as the shared drivers of the three hazard processes.[cite:143][cite:149]

Hazard-response layer

The second layer should define conditional hazard models for wind, surge, and rain.

Wind model

The wind model should consume the storm state and generate local wind speed using the radial wind-field structure defined elsewhere in the specification.[cite:98][cite:99] At minimum, local wind should depend on:

- Distance from storm center.
- Radius of maximum wind.
- Peak sustained wind.
- Motion asymmetry.
- Land interaction.[cite:98][cite:99]

Conceptually:

$$W \sim P_W(\cdot \mid Z)$$

Equation (17)

Surge model

The surge model should be conditional on storm state plus local coastal exposure. Surge is driven not only by wind magnitude but also by storm size, pressure structure, heading, coastal orientation, and local bathymetric or geometric amplifiers.[cite:143][cite:149][cite:150]

A simplified conditional form is:

$$S = g_S(Z, L_{\text{coast}}) + \varepsilon_S$$

Equation (18)

Where L_{coast} represents location-specific coastal modifiers. The first implementation may use a stylized surge proxy rather than a full hydrodynamic model, but it must remain structurally conditional on the storm state rather than merely correlated with wind.[cite:143][cite:149]

Rain model

The rain model should be conditional on storm state plus local terrain or moisture modifiers. Rainfall should depend on at least storm size, translation speed, persistence, land interaction, and optional moisture/orographic proxies.[cite:141][cite:149][cite:150]

A simplified conditional form is:

$$R = g_R(Z, L_{\text{terrain}}) + \varepsilon_R$$

Equation (19)

Where L_{terrain} represents local terrain, coastal, or exposure modifiers. The coding team should note that slow translation speed may materially increase accumulated rainfall, even when peak wind is not at its most extreme.[cite:141][cite:149][cite:150]

Residual dependence layer

After conditioning on the latent storm state, some dependence may remain between wind, surge, and rain, especially in the upper tail. This should be handled with an explicit residual dependence layer, not by forcing the conditional models themselves to absorb all co-movement.[cite:145][cite:151]

The first implementation may use one of the following:

- Tail-aware copula.
- Vine copula for flexible pairwise dependence.
- Conditional extremes model for upper-tail co-exceedance.[cite:145][cite:151]

The residual layer should be weak relative to the shared storm-state layer in ordinary events, but it may become more important in the extreme tail or under specific storm regimes.[cite:145][cite:151]

Regime-specific dependence

Dependence parameters should be allowed to vary by storm regime, because a slow-moving large landfalling storm does not produce the same joint hazard structure as a fast compact offshore storm.[cite:141][cite:143][cite:148]

At minimum, the coding team should support regime-specific dependence profiles for:

- Compact high-wind storms.
- Large slow-moving storms.
- Coastal-grazing storms.
- Recurving offshore storms.[cite:141][cite:143][cite:148]

This requirement is important because compound hazard behavior is event-type dependent and cannot be captured adequately by one global tri-variate dependence model.[cite:141][cite:143][cite:145]

Tail treatment in the joint hazard model

The joint hazard layer must integrate with the tail-event framework already defined in this specification. In particular, the model should recognize that compound tail events may arise from:

- Extreme wind with moderate rain and high surge.
- Moderate-high wind with very high surge and very high rain.
- Long-duration moderate wind with exceptional rainfall accumulation.
- Large storms producing broad but not necessarily record peak winds.[cite:123][cite:141][cite:143][cite:149]

This means tail classification should not rely only on peak wind. It should incorporate multi-hazard exceedance behavior and compound tail combinations.[cite:123][cite:126]

Calibration requirements

Calibration should proceed in stages.

1. Calibrate the latent storm-state model separately.[cite:106][cite:107]
2. Calibrate the wind module against wind-field expectations.[cite:98][cite:99]
3. Calibrate the surge proxy or surge-response function against representative coastal behavior.[cite:143][cite:149]
4. Calibrate the rainfall model against duration, speed, and size behavior.[cite:141][cite:149][cite:150]
5. Estimate residual dependence on standardized residuals after conditioning on storm state.[cite:145][cite:151]
6. Refit by regime if material structural differences are observed.[cite:141][cite:148]

The coding team should not attempt to estimate the full joint tri-variate dependence in one step before the conditional hazard layers are stable. The hierarchy is the primary structural assumption and should be respected during calibration.[cite:145][cite:151]

Validation requirements

Validation should include both marginal and joint behavior.

Required tests include:

- Realism of individual wind, surge, and rain marginals.[cite:141][cite:143][cite:149]
- Realism of pairwise dependence: wind-surge, wind-rain, surge-rain.[cite:141][cite:143][cite:154]
- Realism of tri-variate co-exceedance rates.[cite:145][cite:151]
- Tail stability under Monte Carlo simulation.[cite:78][cite:79]
- Regime sensitivity and explainability under scenario changes.[cite:141][cite:148]

The team should pay particular attention to whether the model reproduces plausible compound flooding situations in which surge and heavy precipitation co-occur under the same storm system.[cite:141][cite:143][cite:146]

Sufficiency statement

For the purposes of this platform, this hierarchical joint hazard design should be regarded as sufficiently sophisticated. It captures the dominant physical driver structure, permits explicit tail modeling, supports modular calibration, and provides an auditable dependence architecture suitable for scientific review and model governance.[cite:145][cite:148][cite:153]

A more elaborate model, such as a fully coupled dynamical coastal-hydrology-atmospheric simulator, may be scientifically richer but is not necessary for the current objective. The current hierarchy is therefore the recommended stopping point for the first production-grade architecture.[cite:141][cite:143][cite:150]